FROM FOAM RUN TO SURFACE PHYSICS

An updated, evidence-led post-processing guide for the cusped-leading-edge airfoil case

OpenFOAM Foundation 13 | shockFluid | Laminar, compressible, transient flow

THE CENTRAL IDEA

A solver run is not an aerodynamic result. Post-processing checks numerical behaviour, converts fields into quantities that answer engineering questions, and tests whether those quantities are trustworthy.

What changed in this edition

The previous guide stopped just before the most important surface work. Since then, we deliberately moved from asking only what happens along a sampling line to asking what the flow is doing on the airfoil itself. We exported the wall patch, inspected pressure and temperature, probed the near-wall flow, calculated wall shear and aerodynamic forces, and checked the calculations against OpenFOAM native function objects.

The workflow followed

Stage	Question answered	Current status
1. Preserve and audit	Can the run and its saved data be reproduced?	Completed for the first startup run
2. Numerical and physical sanity	Did the run progress, and does the freestream make sense?	Checked; accuracy not certified
3. Derived and volume fields	What do Mach, speed, vorticity and field statistics show?	Completed at the saved final time
4. Line profiles	What changes along a chosen fluid transect?	Completed as supporting evidence
5. Surface field inspection	How do pressure and temperature vary over the airfoil?	Completed at the saved final time
6. Surface viscous physics	What are wall shear stress and a preliminary Cf?	Calculated; leading-edge sensitivity remains
7. Integrated loads	What pressure/viscous force and coefficient act on the airfoil?	Independently cross-checked
8. Time and validation	Do results settle and survive mesh/domain/time-step changes?	Not yet completed

Case record updated 02 November 2026. The guide describes the actual case history and results supplied in the working session; suggested next steps are explicitly labelled.

1. Where the case stands now

The geometry and solver setup remain the same as the previous guide. This edition updates the post-processing evidence, not the underlying simulation setup.

Geometry, mesh and domain

Item	Observed state
Case directory	~/Projects/cusped-airfoil/openfoam-case
Solver	OpenFOAM Foundation 13; foamRun with solver shockFluid
Mesh	24,000 hexahedral cells; 48,680 points; 96,340 faces
Boundary patches	aerofoil (wall), inlet, outlet, back (empty), front (empty)
Airfoil wall resolution	220 wall faces; surface VTK contains 440 points and 220 cells
Bounding box	x = -1.25 to 4 m; y = -0.1 to 0 m; z = -2 to 2 m
Mesh quality	max aspect ratio 81.264; max non-orthogonality 87.269 degrees; average 13.951 degrees; 186 faces above 70 degrees; max skewness 3.082
Mesh status	checkMesh reports “Mesh OK”, but the high non-orthogonality is a serious quantitative-accuracy caveat

Flow setup and saved times

Parameter	Current case
Flow model	Laminar, compressible, ideal gas, Sutherland viscosity
Freestream	M_inf = 1.25; T_inf = 300 K; p_inf = 1775.544 Pa
Freestream velocity	U_inf = 433.9 m/s along +X; alpha = 0 degrees
Run controls	deltaT = 1e-7 s; endTime = 2e-5 s; writeInterval = 5e-6 s
Saved times	0, 5e-06, 1e-05, 1.5e-05, 2e-05 s
Latest analysed state	2e-05 s

DO NOT OVERCLAIM

The run completed, but it is still a very short startup calculation. The far-field domain is much smaller than the paper’s reported far field, the boundary conditions are a simplified adaptation, and mesh non-orthogonality remains high. The force agreement shown later verifies the post-processing arithmetic, not the physical validity of the flow solution.

2. The actual post-processing route

This is the chronology we followed, including the deliberate switch toward surface data.

Step	What we did	Why
A. Preserve and inspect	Listed saved times and checked solver output.	Establish exactly which data exist.
B. Check numerical health	Read end time, Courant number, residual behaviour and errors.	Identify numerical failure or suspicious behaviour.
C. Check freestream physics	Calculated sound speed, density, dynamic pressure, viscosity and chord time.	Create independent reference values.
D. Create derived fields	Calculated Mach number; inspected speed, vorticity and gradients.	Turn primitive fields into more meaningful diagnostics.
E. Inspect volume fields	Exported VTK and viewed spatial fields.	Locate where the flow is disturbed.
F. Sample a line	Used graphUniform and plotted p, T, Mach and velocity components.	Get quantitative values along a defined transect.
G. Pivot to the wall	Exported patch data and focused on the aerofoil surface.	Answer where loading and shear occur over the body itself.
H. Probe near the wall	Sampled fluid points at known normal offsets from the upper surface.	Understand how wall disturbances decay into the flow.
I. Calculate wall shear	Read wallShearStress and formed a magnitude-based preliminary Cf.	Quantify viscous loading locally.
J. Integrate forces	Computed pressure and viscous loads and coefficients.	Convert local surface data into whole-body loads.
K. Cross-check	Compared Python integration with forcesCompressible and forceCoeffsCompressible.	Independently verify integration and normalisation.

THE SURFACE-FIRST DECISION

The original request was not to keep making more line plots. We therefore made the wall patch the main object of study. Line sampling was retained as a supporting diagnostic, but the central analysis became surface pressure, temperature, shear stress, and integrated loads.

Keep the distinction clear

A line sample answers: “What does this field do along this particular path?” A wall-surface extraction answers: “How does this field vary from face to face over the airfoil?” A near-wall probe answers: “How does the fluid state change as I move away from the wall?” These are related, but they are not interchangeable measurements.

3. Numerical health and freestream sanity

These checks remain the first filter before interpreting any aerodynamic feature.

Run health

Log item	Observed result	Interpretation
End	Reached 2e-05 s and exited normally.	The solver advanced to the configured end time.
Courant number	Mean about 0.001952; maximum about 0.484876 at final post-process.	The maximum was below 0.5 in the reported run; this is a stability/accuracy indicator, not an accuracy certificate.
Residuals	Final linear-solver residuals were very small.	The linear systems were solved tightly; this does not prove the transient flow is settled.
Errors	No fatal exit reported in the supplied run log.	No obvious solver crash in this run.

Useful audit commands

```
cd ~/Projects/cusped-airfoil/openfoam-case
foamListTimes
ls -lh 2e-05/
grep -E "Time =|Courant Number|FOAM FATAL|^End" log.foamRun | tail -n 50
```

Residuals are remaining algebraic imbalances in the discretised equation solves. They are not direct measurements of physical error. For a transient case, inspect the trends across time steps, the actual fields and the force history. Look for non-finite data, negative pressure/temperature/density, unexpected drift, or sudden spikes.

Freestream reference calculations

Quantity	Relation / assumption	Reference value
Specific gas constant	R = R_u / molecular weight	287.042 J/(kg K)
Sound speed	a_inf = sqrt(gamma R T_inf)	about 347.2 m/s
Velocity	U_inf = M_inf a_inf	433.9 m/s
Density	rho_inf = p_inf/(R T_inf)	0.02061885 kg/m^3
Dynamic pressure	q_inf = 0.5 rho_inf U_inf^2	1940.9477 Pa
Dynamic viscosity	Sutherland law at 300 K	about 1.85e-5 Pa s
Reynolds number	rho_inf U_inf c / mu_inf; c = 1 m	about 4.85e5
Chord travel time	t_c = c/U_inf	about 2.30e-3 s

RUN DURATION IN CONTEXT

The current end time is 2.0e-5 s, only about 0.0087 chord-travel times (0.87%). It is an early transient snapshot, not an established steady or time-averaged aerodynamic solution.

4. Derived fields and volume-level inspection

The purpose here was to understand the flow before concentrating on the airfoil wall.

Mach number

```
foamPostProcess -list | grep -i MachNo
foamPostProcess -solver shockFluid -latestTime -func MachNo 2>&1 | tee log.postMach
find 2e-05 -maxdepth 1 -type f -printf "%f\n" | sort
```

Mach number is local speed divided by local speed of sound. The `MachNo` function object generated the `Ma` field at the latest saved time. Upstream freestream should be close to 1.25. A shock should be assessed through a consistent combination of Mach, pressure and temperature changes, not through a colour contour alone.

Field statistics at t = 2e-05 s

Field	Minimum	Maximum	Mean	Median
p (Pa)	977.077	2183.41	1776.4	1775.54
T (K)	255.148	358.499	300.091	300
Ma (-)	0.302456	1.36951	1.24904	1.25011

The median Mach number is close to the target freestream value. Domain-wide minima and maxima include the airfoil vicinity and should not be mistaken for freestream values. The pressure and temperature extrema require spatial context before assigning a physical cause.

Useful fields

Field / operation	What it adds	Caution
mag(U)	Velocity magnitude.	Does not show direction or account for local sound speed.
vorticity	Curl of velocity; highlights rotational/shear regions.	Gradient-sensitive; large peaks can be mesh-sensitive.
grad(p)	Pressure-gradient vector.	Strongly affected by shock resolution and mesh skewness.
wallShearStress	Wall shear vector on wall patches.	Requires correct wall patch and solver/model context.
Ma	Local Mach number.	Must be interpreted alongside p, T and spatial location.

```
foamToVTK -latestTime -ascii
```

The VTK export created volume and patch files for spatial inspection. The volume file can be used for a mid-plane slice; the patch-specific wall file is more direct for face-by-face surface work. The two datasets should not be confused.

5. Line sampling: useful, but no longer the centrepiece

We did complete a line profile. It helped us see where to focus, but it did not replace direct surface extraction.

What was sampled

The case profile lies in the X-Z plane and the mesh is extruded in Y from -0.1 to 0 m. The mid-plane used in the diagnostics was y = -0.05 m. The graphUniform setup specified a line, number of points and requested fields. Its saved output was:

```
system/graphUniform
postProcessing/graphUniform/2e-05/line.xy
```

The sampled columns contained x-coordinate, pressure, temperature and velocity components. The plotted panels showed pressure, temperature, Mach and vertical velocity along the chosen transect. Far from the airfoil, the values stayed close to the freestream state; changes became apparent near the airfoil region around x approximately 0.65-0.80 m.

What the profile suggested

Observed feature	What it suggested	What it did not prove
Pressure and temperature change near x ~ 0.65-0.80 m.	The flow has a strong local disturbance in this region.	A shock location or shock strength by itself.
Mach peak around 1.31 and dip around 1.24 in the plotted line.	Acceleration and deceleration occur along the selected path.	A complete 2D description of the flow.
Vertical velocity changes near the airfoil.	The flow turns in the vicinity of the profile.	The full near-wall direction field or separation state.

WHY WE MOVED TO THE SURFACE

The line is only one path through the domain. It can miss the variation from leading edge to trailing edge and cannot directly supply a face-resolved wall-load distribution. The next question was therefore: what pressure, temperature and viscous stress exist on the `aerofoil` patch?

6. Export and inspect the airfoil wall patch

This is where the workflow turns from volume diagnostics to surface physics.

Export the latest fields

```
cd ~/Projects/cusped-airfoil/openfoam-case
foamToVTK -latestTime -ascii
```

The export includes the volume fields and patch fields. In this case, the preferred face-resolved aerofoil dataset was:

```
VTK/aerofoil/aerofoil_200.vtk
```

Dataset	Contents / use
VTK/openfoam-case_200.vtk	Volume mesh with 24,000 cells and 48,680 points; useful for spatial fields and slices.
VTK/aerofoil/aerofoil_200.vtk	Airfoil patch: 220 cells and 440 points; contains both cell and point arrays, including p, T, U, Ma, rho, wallShearStress and vorticity.
postProcessing/surfaces/2e-05/aerofoilWall.vtk	Surface-sampling output with geometry and point fields; useful, but not the preferred file for face-centred patch values in this analysis.

Why prefer the patch cell data here?

The wall patch is made of faces. Face-centred/cell data represent the field on each wall face and are the natural basis for a surface integral. Point data are interpolated to mesh vertices and can be useful for smooth visualisation, but they are not the same sampling representation. When extracting data, record whether the values are cell data or point data.

DO NOT MIX REPRESENTATIONS

Use the patch geometry, face centres, face areas and cell/face field values consistently when integrating pressure or viscous traction. A visually smooth point-data contour is not itself a force integration.

What was present on the patch

The extracted patch had 220 surface faces. Available cell arrays included `patchID`, `mag(U)`, `Ma`, `rho`, `p`, `T`, `wallShearStress`, `U` and `vorticity`. This made it possible to inspect the wall state and then integrate its loads.

7. Surface pressure and temperature

The focus now is the wall distribution, not a line through the surrounding fluid.

Observed surface feature near x/c = 0.754

The face-resolved surface data showed an abrupt change around x/c = 0.754202. The recorded adjacent values were:

Quantity	Before the jump	After the jump	Change
Static pressure p	1870.44 Pa	1345.60 Pa	-524.84 Pa
Temperature T	304.727 K	278.551 K	-26.176 K

The upper and lower surface data showed nearly symmetric changes. The feature occurs near a geometric change in the aft portion of the profile: the upper-surface slope becomes more negative after approximately x = 0.758 m.

INTERPRETATION BOUNDARY

This is a strong surface-field change. It may be associated with local expansion/acceleration or a numerical/geometry-resolution feature. Do not label it a shock from this wall jump alone: a shock normally requires the expected compression signature and spatial topology in the surrounding flow.

How to verify the feature

- Map the local pressure and Mach fields in the volume, not only along the wall.
- Inspect several cells normal to the wall and check whether the jump is resolved over a physically plausible distance.
- Refine the local mesh and compare the jump location and magnitude.
- Check that the CAD profile and STL triangulation represent the intended cusp and aft geometry without unintended facets.

Surface field versus near-wall probe

The wall patch values above are wall-face values. The probe data in the next section are fluid values at specified distances away from the surface. They should be reported separately, even when they show a consistent trend.

8. Near-wall probes: how the disturbance decays

We sampled fluid points at known normal offsets from the upper surface at selected chordwise locations.

The probe set used x/c stations 0.72, 0.74, 0.76, 0.80, 0.85, 0.90 and 0.98, and normal offsets 0.002, 0.005, 0.010, 0.020, 0.040, 0.080 and 0.120 m. The earlier NaN issue was traced to unsuccessful point-to-cell lookup; after correcting the probe selection, all 49 of 49 points were found in cells.

Selected upper-side results at 2 mm from the wall

x/c	Ma	p (Pa)	T (K)	Ux (m/s)	Uz (m/s)
0.74	1.2789	1870.44	304.727	447.28	+9.1325
0.76	1.2449	1333.52	277.979	414.447	-35.164
0.80	1.2411	1287.77	275.369	410.761	-40.096
0.90	1.2205	1147.92	266.993	396.267	-51.864
0.98	1.1917	1040.23	260.03	380.302	-60.511

At x/c = 0.76 and 0.80, the near-wall fluid is slower in the streamwise direction, has lower pressure and temperature than the point at x/c = 0.74, and has a stronger negative vertical velocity. Moving away from the wall, the sampled values return toward freestream conditions; at several stations they are close to freestream by about 0.02 m offset.

WHAT THESE PROBES TELL US

There is a strong near-wall disturbance and flow turning over the aft portion of the upper surface. This supports the surface-field observation, but the probe results alone do not identify a shock, separation point, or validated boundary-layer thickness.

These are sampled fluid-cell/interpolated values at 2 mm normal offset, not wall-face values. The listed set is the upper side only.

9. Wall shear stress and a preliminary skin-friction coefficient

This is the local viscous-loading analysis on the `aerofoil` wall patch.

Wall shear stress field

The `wallShearStress` output contained 220 wall-face vectors. Its dimensions were [1 -1 -2 0 0 0 0], i.e. pressure units (Pa). We examined the vector magnitude and mapped it by wall-face location.

Quantity	Observed value
Minimum |tau_w|	5.084997 Pa
Maximum |tau_w|	92.457946 Pa
Location of global maximum	Near leading edge, x about 0.001881345 m; y = -0.05 m; z about -1.869e-6 m
Upper-side maximum	about 92.3798 Pa
Lower-side maximum	about 92.4579 Pa
Spanwise component	Zero for this 2D-style field
Maximum in-plane normal component	about 5.3988 Pa

Magnitude-based Cf

For a first comparison, we formed a magnitude-based coefficient using the freestream dynamic pressure:

PRELIMINARY DEFINITION

Cf_mag = |tau_w| / q_inf, with q_inf = 1940.9477 Pa for this case.

Statistic	Cf_mag
Minimum	0.00261985
Maximum	0.04763547

The large leading-edge maximum is likely sensitive to the local mesh and the sharp geometry. It should not be treated as a converged local skin-friction peak. The current maximum non-orthogonality is 87.27 degrees.

Selected surface stations

x/c	|tau_w| (Pa)	Cf_mag
0.9133	5.909	0.00304
0.7878	6.336	0.00326
0.7406	7.334	0.00378
0.4971	7.885	0.00406
0.3185	8.472	0.00437
0.1867	9.551	0.00492
0.08934	13.089	0.00674
0.01757	39.566	0.02038

WHAT IS NOT FINISHED

A proper signed Cf distribution requires projecting the wall-shear vector onto a consistently defined local tangent direction. The values above are magnitude-based and should be labelled that way. They are not a signed upper/lower skin-friction curve.

10. Surface geometry, normals and integration area

Before integrating wall loads, the face orientation and reference area must be internally consistent.

Surface geometry checks

Check	Observed result
Extrusion depth	0.1 m (y from -0.1 to 0 m)
Reference chord	1.0 m
Reference area	A_ref = chord x extrusion depth = 0.1 m^2
Upper wall area	0.10068413 m^2
Lower wall area	0.10068413 m^2
Total wall area	0.20136826 m^2
Upper mean normal	approximately (-0.008839, 0, -0.99308)
Lower mean normal	approximately (-0.008839, 0, +0.99308)
Closed-surface area-vector sum	approximately (8.44e-11, 0, 4.41e-17) m^2

The opposing upper/lower normal directions and near-zero total area-vector sum are consistent with a closed airfoil surface. This check matters because pressure traction depends on the face normal. The reference area is not the physical total wetted area: it is chord times the extrusion depth, used to nondimensionalise the 2D-style force result.

Force and coefficient equations

Quantity	Definition for this case
Pressure force	Integrate pressure traction over the airfoil patch using face area vectors.
Viscous force	Integrate the viscous traction over the same patch.
Total force	F_total = F_pressure + F_viscous
Drag coefficient	Cd = D / (q_inf A_ref)
Lift coefficient	Cl = L / (q_inf A_ref)
Pitching moment coefficient	Cm = M_pitch / (q_inf A_ref c_ref)

Axes are tied to the model: flow is +X, lift is +Z, extrusion/pitch axis is Y. The `CofR` used for the moment calculation was (0.25, -0.05, 0) m.

11. OpenFOAM force function objects

We configured both coefficient and dimensional-force calculations in Foundation v13.

system/forceCoeffsCompressible

```
patches         (aerofoil);
magUInf         433.9;
rhoInf          0.02061885;
lRef            1.0;
Aref            0.1;
CofR            (0.25 -0.05 0);
dragDir         (1 0 0);
liftDir         (0 0 1);
pitchAxis       (0 1 0);
pRef            1775.544;
log             yes;
#includeEtc "caseDicts/functions/forces/forceCoeffs.cfg"
```

system/forcesCompressible

```
patches         (aerofoil);
CofR            (0.25 -0.05 0);
pitchAxis       (0 1 0);
log             yes;
#includeEtc "caseDicts/functions/forces/forces.cfg"
```

system/functions

```
FoamFile
{
    format      ascii;
    class       dictionary;
    location    "system";
    object      functions;
}

#includeFunc forceCoeffsCompressible
#includeFunc forcesCompressible
```

Run the two function objects

```
foamPostProcess \
    -dict system/functions \
    -solver shockFluid \
    -latestTime
```

The solver option is important because these compressible post-processing objects need the solver/thermophysical context. The `forcesCompressible` function reports dimensional pressure and viscous forces; `forceCoeffsCompressible` reports nondimensional Cd, Cl and Cm. The Foundation v13 guide documents these as distinct function objects.

WHY RUN BOTH?

A coefficient by itself is convenient but hides the dimensional load decomposition. The dimensional-force function exposes pressure and viscous contributions separately, making it easier to diagnose how the total is formed.

12. Dimensional forces: pressure plus viscous

The native OpenFOAM result at t = 2e-05 s was:

Contribution	Fx (N)	Fz (N)
Pressure	9.178529	-0.000146746
Viscous	1.909740	+0.000015988
Total	11.088269	-0.000130758

The detailed OpenFOAM console output also reported the moments about the configured centre of rotation:

Contribution	My (N m)
Pressure	-3.640959e-05
Viscous	+3.967361e-06
Total	-3.244223e-05

Saved output file

```
postProcessing/forcesCompressible/2e-05/forces.dat
```

The saved line contains the pressure and viscous force vectors and their moments. The console log and file both show the pressure and viscous force components separately. This is the dimensional-load result used for the independent comparison.

Python integration result

Contribution	Python Fx (N)	OpenFOAM Fx (N)
Pressure	9.178528612	9.178529
Viscous	1.909739899	1.909740
Total	11.088268512	11.088269

Python total lift was -0.000130760 N; OpenFOAM reported -0.000130758 N. The difference is below the precision shown in the rounded OpenFOAM output.

WHAT THIS VALIDATES

The Python face-area/traction integration and OpenFOAM native dimensional-force integration agree to the displayed precision. This verifies the numerical consistency of the surface integration for this saved snapshot.

13. Force coefficients and independent cross-check

The dimensional-force agreement is complemented by the coefficient calculation.

OpenFOAM forceCoeffsCompressible output

Coefficient	OpenFOAM value
Cd	0.05712799 (console rounded: 0.057128)
Cl	-6.736817e-07
Cm	-1.671459e-07
Cl(f)	-5.03987e-07
Cl(r)	-1.69695e-07

The saved coefficient file was `postProcessing/forceCoeffsCompressible/2e-05/forceCoeffs.dat`. Its reported total coefficients are consistent with the direct force calculation after reference-area and dynamic-pressure normalisation.

Python versus OpenFOAM

Quantity	Python	OpenFOAM	Agreement
Cd	0.057128115	0.05712799	Difference about 1.25e-7
Cl	about -6.74e-7	-6.736817e-7	Matches at displayed precision
Total Fx	11.088268512 N	11.088269 N	Rounding only
Total Fz	-0.000130760 N	-0.000130758 N	Rounding only

What we can now say

- The wall face integration, force directions and reference normalisation are internally consistent between two independent routes.
- The pressure contribution is about 82.8% of the total drag; the viscous contribution is about 17.2%.
- The airfoil is symmetric and the prescribed angle of attack is zero; the near-zero net lift is consistent with that setup.

IMPORTANT TERMINOLOGY

The 82.8% pressure contribution is pressure drag. Do not automatically relabel the entire contribution as wave drag. A wave-drag interpretation needs a validated shock/pressure-field analysis.

14. How to interpret the current aerodynamic result

A successful cross-check is a post-processing milestone. It is not yet a validated aerodynamic prediction.

Snapshot interpretation at t = 2e-05 s

Result	Value	Interpretation / limitation
Pressure drag	9.17853 N	Dominant part of the integrated drag at this instant.
Viscous drag	1.90974 N	About 17.2% of total drag at this instant.
Total drag	11.08827 N	Corresponds to Cd about 0.057128 with A_ref = 0.1 m^2.
Total lift	-1.31e-4 N	Practically zero for the symmetric zero-incidence setup.
Pitch moment coefficient	-1.67146e-7	Very small for the current reference centre and snapshot.

Evidence that is strong

- The case runs to the configured end time without a fatal error.
- The freestream median Mach is close to the target 1.25.
- Wall patch extraction provides a face-resolved dataset for p, T, U, Ma and wall shear.
- Independent Python and OpenFOAM integrations agree for both dimensional forces and nondimensional coefficients.

Evidence that is still weak or incomplete

- The mesh has severe local non-orthogonality warnings, particularly consequential for gradients and wall shear near the cusp.
- The far-field domain is substantially smaller than the paper's reported domain; boundary reflections/influence are not excluded.
- The simulation duration is only about 0.87% of one chord travel time.
- The abrupt wall pressure/temperature change needs local volume-field inspection and refinement before assigning a definitive flow feature.
- The current Cf curve is magnitude-based; a signed local-tangent projection remains to be completed.

CORRECT CONCLUSION

We have a reproducible post-processing chain and a verified force-integration implementation for the current numerical snapshot. We do not yet have a grid-independent, time-converged, paper-validated aerodynamic result.

15. Next work: from one snapshot to a defensible result

The next tasks continue the same workflow; they do not invalidate what has already been completed.

A. Build force histories from all saved times

The current force functions were executed with `-latestTime`, so the result shown above is for 2e-05 s only. The saved states are 5e-06, 1e-05, 1.5e-05 and 2e-05 s. Run the configured functions over those times using the time-selection option supported by the local Foundation v13 utility; verify its syntax with `foamPostProcess -help` before execution.

```
foamPostProcess -help | less

# After confirming the local -time syntax, process the saved times
foamPostProcess -dict system/functions -solver shockFluid -time \
    "5e-06,1e-05,1.5e-05,2e-05"
```

Plot Cd(t), Cl(t) and Cm(t). Four snapshots are only an initial history, but they will show whether the load is drifting rapidly or changing slowly. A longer physical run will still be needed.

B. Complete surface Cf properly

- Define a local tangent on each face using the ordered airfoil profile or a consistent geometric derivative.
- Project the wall-shear vector onto that tangent and retain a clearly defined sign convention.
- Plot upper and lower signed Cf against x/c; compare with the existing magnitude-based Cf only as a diagnostic.

C. Resolve the pressure/temperature jump

- Inspect the volume fields in the region around x/c = 0.75-0.80, including several wall-normal cells.
- Refine the local geometry and mesh; compare wall p/T and probe profiles.
- Check whether the feature persists when the mesh is improved and the time step/domain are changed.

D. Validation plan

Test	Minimum comparison	Monitor
Mesh sensitivity	At least two or three systematically refined grids.	Cp(x/c), Cf(x/c), Cd, Cl and local flow topology.
Domain sensitivity	Enlarge the far field and compare with current domain.	Shock/pressure pattern, wall loads and boundary influence.
Time-step sensitivity	Reduce deltaT while holding other inputs fixed.	Fields and force histories.
Duration sensitivity	Run for a physically meaningful number of chord times.	Transient development, settled mean or repeatable oscillations.
Boundary-condition sensitivity	Review inlet/outlet/far-field treatment for the supersonic arrangement.	Freestream recovery and reflected disturbances.
Paper comparison	Match geometry, alpha, freestream, reference values and reported conditions.	Only compare like-for-like quantities.

16. Reproducibility and reporting checklist

Every reported figure or number should be traceable to its data source and method.

Keep the working record

```
cd ~/Projects/cusped-airfoil/openfoam-case
mkdir -p notes logs
cp log.foamRun logs/log.foamRun.first-startup
foamListTimes > notes/time-directories.txt
checkMesh > logs/checkMesh.final.txt 2>&1
find postProcessing -type f -print | sort > notes/postProcessing-files.txt
```

Before substantial changes, preserve the working case and make one controlled change at a time. Keep the generated VTK data, force files, extraction scripts, line data and plots linked to the exact time directory used.

For every figure or table, record five things

Record	What to write down
Question	What engineering question is this figure meant to answer?
Method	Which field, function object, patch, sampling line/probe, interpolation representation and time were used?
Result	Report values with units, axes and extrema or characteristic values.
Interpretation	Separate the observation from the physical explanation.
Limitation	State mesh, domain, time-step, boundary-condition or sampling uncertainties.

Updated milestone statement

PROFESSOR-READY SUMMARY

I adapted the OpenFOAM Foundation 13 `shockFluid` workflow to a cusped-leading-edge airfoil and completed an initial transient laminar compressible run. Post-processing progressed from solver and freestream checks to derived Mach/volume fields, a supporting line profile, direct face-resolved extraction of the `aerofoil` patch, near-wall probes, wall-shear analysis and integrated pressure/viscous loads. The native OpenFOAM `forcesCompressible` and `forceCoeffsCompressible` results agree with an independent Python surface integration to the displayed precision. The result is still an early startup snapshot: mesh non-orthogonality, domain extent, simulation duration, wall-Cf sign convention and mesh/time/domain sensitivity remain to be addressed before quantitative comparison with the paper.

17. References and case-specific notes

The workflow and function-object terminology follow the OpenFOAM Foundation v13 documentation. The numerical values in the case tables come from the supplied run history and the post-processing results recorded during this project.

OpenFOAM documentation

[1] CFD Direct / OpenFOAM Foundation, OpenFOAM v13 User Guide, Chapter 7: Post-processing. https://doc.cfd.direct/openfoam/user-guide-v13/postprocessing

[2] OpenFOAM v13 User Guide, 7.2: Post-processing command-line interface. https://doc.cfd.direct/openfoam/user-guide-v13/post-processing-cli

[3] OpenFOAM v13 User Guide, 7.3: Post-processing functionality (MachNo, wallShearStress, graphUniform, forcesCompressible, forceCoeffsCompressible, probes and surface fields). https://doc.cfd.direct/openfoam/user-guide-v13/post-processing-functionality

[4] OpenFOAM Foundation v13 source guide, MachNo function object. https://cpp.openfoam.org/v13/classFoam_1_1functionObjects_1_1MachNo.html

[5] Local Foundation v13 mesh reference: $FOAM_TUTORIALS/fluid/aerofoilNACA0012.

[6] Local Foundation v13 solver reference: $FOAM_TUTORIALS/shockFluid/diffuserIntake.

[7] Akram, S. and Hasan, N. (2024), “The dynamics of supersonic flow past a new cusped leading edge airfoil”, Progress in Computational Fluid Dynamics, 24(1), 1–28. DOI: 10.1504/PCFD.2022.10052925.

Interpretation notes to preserve

- A wall-face value and a near-wall probe value are different measurements. Label them accordingly.
- Pressure coefficient is not pressure in pascals. Use Cp = (p_wall - p_inf)/q_inf and state the reference state and x/c convention.
- The current guide reports a magnitude-based Cf, not signed Cf.
- The current force coefficients are for the saved t = 2e-05 s snapshot, not a converged or time-averaged result.
- The pressure-force share is not automatically equal to wave drag.

CURRENT STOPPING POINT

Surface post-processing and force cross-validation are complete for the saved final time. The next active task is to establish force histories and then improve the numerical model enough for mesh/domain/time-step sensitivity testing.